Enum:

```
package shoemanagement;
public enum shoe_type {
FORMAL,CASUAL,SPORTS
}
```

Shoe:

```
package shoemanagement;

public class shoe {

public static int autoid;

int shoe_id;

String name;

String Brand;

int rating;

double price;

boolean AvailableInGallery;

shoe_type type;

static {

autoid = 0;

}

public shoe(String name, String brand, int rating, double price, boolean
availableInGallery,

shoemanagement.shoe_type shoe_type) {

super();

this.shoe_id = ++autoid;

this.name = name;

Brand = brand;

this.rating = rating;

this.price = price;

AvailableInGallery = availableInGallery;

this.type = shoe_type;

}

public int getShoe_id() {

return shoe_id;
```

```
}
public String getName() {
return name;
}
public void setName(String name) {
this.name = name;
}
public String getBrand() {
return Brand;
}
public void setBrand(String brand) {
Brand = brand;
}
public int getRating() {
return rating;
}
public void setRating(int rating) {
this.rating = rating;
}
public double getPrice() {
return price;
}
public void setPrice(double price) {
this.price = price;
}
public boolean isAvailableInGallery() {
return AvailableInGallery;
}
public void setAvailableInGallery(boolean availableInGallery) {
AvailableInGallery = availableInGallery;
}
```

```
public shoe_type getType() {

return type;

}

public void setType(shoe_type type) {

this.type = type;

}

public static int getAutoid() {

return autoid;

}

@Override

public String toString() {

return "shoe [shoe_id=" + shoe_id + ", name=" + name + ", Brand=" + Brand + ",
rating=" + rating + ", price="

+ price + ", AvailableInGallery=" + AvailableInGallery + ", shoe_type=" + type +
"]";

}

}
```

ShoeException

```
package shoevalidation;

public class ShoeException extends Exception {

public ShoeException(String message) {
super(message);
}

}
```

ShoeValidation

```
package shoevalidation;
import shoemanagement.shoe_type;
public class ShoeValidation {
public static void ValidateRating(int rating) throws ShoeException {
if (rating < 1 || rating > 10) {
throw new ShoeException("Invalid rating ,rating must be 1 and 10");
}
}
public static void validatenamelength(String name) throws ShoeException {
if (name.length() < 3 || name.length() > 20) {
throw new ShoeException("Invaid shoe name,name must be beetween 3 and 20");
}
```

```
}
public static void validateshoetype(String inputType) throws ShoeException {
boolean isvalid = false;
for (shoe_type type : shoe_type.values()) {
if (type.name().equalsIgnoreCase(inputType)) {
isvalid = true;
break;
}
}
if (!isvalid) {
throw new ShoeException("Invalid shoe type." + inputType);
}
}
}
```

ShoeUtility

```
package Utill;
import java.util.ArrayList;
import java.util.Comparator;
import java.util.List;
import shoemanagement.shoe;
import shoemanagement.shoe_type;
import shoevalidation.ShoeException;
import shoevalidation.ShoeValidation;
public class shoeutility {
private List<shoe> shoeList = new ArrayList<>();
public String addShoe(String name, String brand, int rating, double price, boolean
InAvailableGallery,
String type) {
try {
ShoeValidation.validatenamelength(name);
ShoeValidation.ValidateRating(rating);
ShoeValidation.validateshoetype(type);
for (shoe s : shoeList) {
if (s.getName().equalsIgnoreCase(name)) {
return "Shoe with this name already exists";
}
}
shoe_type shoeTypeEnum = shoe_type.valueOf(type.toUpperCase());
shoe Newshoe = new shoe(name, brand, rating, price, InAvailableGallery,
shoeTypeEnum);
shoeList.add(Newshoe);
return "Shoe added successfully !id:" + Newshoe.getShoe_id();
} catch (ShoeException e) {
return "validation error: " + e.getMessage();
} catch (IllegalArgumentException e) {
return "Invalid shoe type  ";
} catch (Exception e) {
return "unexcpected Error" + e.getMessage();
}
}
public void displayALLunsorted() {
if (shoeList.isEmpty()) {
System.out.println("Shoelist is empty");
} else {
shoeList.forEach(System.out::println);
}
```

```
}
public void displaysortedbyid() {
if (shoeList.isEmpty()) {
System.out.println("ShoeList is empty");
} else {
shoeList.stream().sorted(Comparator.comparingInt(shoe::getShoe_id)).forEach(System
.out::println);
}
}
public void searchbyprice() {

shoeList.stream().max(Comparator.comparingDouble(shoe::getPrice)).ifPresentOrElse(
System.out::println,
() -> System.out.println("No shoes available"));
}
public void removenotavailable() {
boolean removed = shoeList.removeIf(s -> !s.isAvailableInGallery());
if (removed) {
System.out.println(" unavailable shoes removed");
} else {
System.out.println("All shoe are currantly available");
}
}
public void updatepricebybrand(String brand, double newprice) {
if (shoeList.isEmpty()) {
System.out.println("List is empty");
return;
}
boolean updated = false;
for (shoe s : shoeList) {
if (s.getBrand().equalsIgnoreCase(brand)) {
double oldprice = s.getPrice();
s.setPrice(newprice);
updated = true;
System.out.println("Updated price for shoe: " + s.getName() + "| old price: " +
oldprice
+ "| new price: " + newprice);
}
}
if (!updated) {
System.out.println("No shoe found for this brand");
}
}
public void sortbyprice() {
if (shoeList.isEmpty()) {
System.out.println("List is empty");
}
shoeList.stream().sorted(Comparator.comparingDouble(shoe::getPrice)).forEach(Syste
m.out::println);
}
}
```

Tester

```
package tester;
import java.util.Scanner;
import Utill.shoeutility;
public class test {
public static void main(String[] args) {
Scanner sc = new Scanner(System.in);
```

```java
shoeutility su = new shoeutility();
int choice = 0;
do {
System.out.println("------Menu------");
System.out.println("1.Add a new shoe record to collection");
System.out.println("2.display all shoe details ");
System.out.println("3.display all shoe details sorted by id");
System.out.println("4.Search most expensive shoe in gallery");
System.out.println("5.remove shoes that are not available in gallary");
System.out.println("6.Update shoe price based on brand");
System.out.println("7.sort shoe data as per price in descending order");
System.out.println("8.Exit ");
System.out.println("Enter choice");
choice = sc.nextInt();
switch (choice) {
case 1:
System.out.println("Enter name:");
String name = sc.next();
System.out.println("Enter brand:");
String brand = sc.next();
System.out.println("Enter rating(1-5):");
int rating = sc.nextInt();
System.out.println("Enter price:");
double price = sc.nextDouble();
System.out.println("Is shoe available in gallery?(true/false): ");
boolean ia = sc.nextBoolean();
System.out.println("Enter type");
String type = sc.next();
System.out.println(su.addShoe(name, brand, rating, price, ia, type));
break;
case 2:
su.displayALLunsorted();
break;
case 3:
su.displaysortedbyid();
break;
case 4:
su.searchbyprice();
break;
case 5:
su.removenotavailable();
break;
case 6:
System.out.println("Enter brand name:");
String br = sc.next();
System.out.println("Enter new price:");
double newprice = sc.nextDouble();
su.updatepricebybrand(br, newprice);
break;
case 7:
su.sortbyprice();
break;
case 8:
System.out.println("Exiting the program.....");
break;
}
} while (choice != 8);
sc.close();
}
```

}